ClipSAT : Math for AI & Data Science [ANSWER KEY]

Unit 1.01: Vectors & Embeddings in AI

Q1. Compute the Manhattan L_1 norm and Euclidean L_2 norm of the word vector mathbf{w} = [10, -15]^T .

[ ] A: |mathbf{w}|_1 = sqrt{325}, ; |mathbf
[ ] B: |mathbf{w}|_1 = -5, ; |mathbf{w}|_2
[ ] C: |mathbf{w}|_1 = 25, ; |mathbf{w}|_2
[x] D: |mathbf{w}|_1 = 25, ; |mathbf{w}|_2

Solution: L_1 norm is the sum of absolute values: |10| + |-15| = 25 . L_2 norm is the square root of sum of squares: sqrt{10^2 + -15^2} = sqrt{100 + 225} = sqrt{325} .

Q2. Two document embeddings mathbf{a} = [1, 2]^T and mathbf{b} = [2, 1]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{5} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[x] C: 0.800
[ ] D: 0.500

Solution: Dot product mathbf{a} cdot mathbf{b} = 12 + 21 = 4 . Product of norms |mathbf{a}| |mathbf{b}| = 5 . Cosine similarity = frac{4}{5} = 0.800 .

Q3. Two document embeddings mathbf{a} = [2, 4]^T and mathbf{b} = [4, 2]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{20} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[x] C: 0.800
[ ] D: 0.500

Solution: Dot product mathbf{a} cdot mathbf{b} = 24 + 42 = 16 . Product of norms |mathbf{a}| |mathbf{b}| = 20 . Cosine similarity = frac{16}{20} = 0.800 .

Q4. Two document embeddings mathbf{a} = [3, 6]^T and mathbf{b} = [6, 3]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{45} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[x] C: 0.800
[ ] D: 0.500

Solution: Dot product mathbf{a} cdot mathbf{b} = 36 + 63 = 36 . Product of norms |mathbf{a}| |mathbf{b}| = 45 . Cosine similarity = frac{36}{45} = 0.800 .

Q5. Two document embeddings mathbf{a} = [4, 8]^T and mathbf{b} = [8, 4]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{80} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[x] C: 0.800
[ ] D: 0.500

Solution: Dot product mathbf{a} cdot mathbf{b} = 48 + 84 = 64 . Product of norms |mathbf{a}| |mathbf{b}| = 80 . Cosine similarity = frac{64}{80} = 0.800 .

Unit 1.01: Vectors & Embeddings in AI

Q6. Two document embeddings mathbf{a} = [5, 10]^T and mathbf{b} = [10, 5]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{125} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[x] C: 0.800
[ ] D: 0.500

Solution: Dot product mathbf{a} cdot mathbf{b} = 510 + 105 = 100 . Product of norms |mathbf{a}| |mathbf{b}| = 125 . Cosine similarity = frac{100}{125} = 0.800 .

Q7. Two document embeddings mathbf{a} = [6, 12]^T and mathbf{b} = [12, 6]^T are compared. If |mathbf{a}| = |mathbf{b}| = sqrt{180} , calculate their exact cosine similarity.

[ ] A: 1.000
[ ] B: 0.640
[x] C: 0.800
[ ] D: 0.500

Solution: Dot product mathbf{a} cdot mathbf{b} = 612 + 126 = 144 . Product of norms |mathbf{a}| |mathbf{b}| = 180 . Cosine similarity = frac{144}{180} = 0.800 .

Q8. The coordinate diagram shows the parabolic curve f_1x = x^2 - 6x + 20 in the domain of Vectors & Embeddings in AI. By completing the square f_1x = x - h^2 + k , find the sum k + f_15 .

[x] A: 26
[ ] B: 29
[ ] C: 15
[ ] D: 24

Solution: **Mark Scheme & Examiner Solution Vectors & Embeddings in AI:** 1. Complete the square: x^2 - 6x + 20 = x - 3^2 - 9 + 20 = x - 3^2 + 11 , giving minimum value k = 11 . **[M1]** 2. Evaluate at x = 5 : f_15 = 2^2 + 11 = 15 . **[M1]** 3. Sum the two quantities: k + f_15 = 11 + 15 = 26 . **[A1]** - **Final Verified Answer:** 26 .